

Here $\boxed{x=0}$
about origin

Moment generating function.

①

Let x is a Random variable then function

$E(e^{tx})$ denoted by $M(t)$ is called Moment-generating function of variable x , where t is a constant. Since function $M(t)$ generates Moments, Hence is called Moment generating function.

$$\begin{aligned}M_x(t) &= E(e^{tx}) = E\left[1 + tx + \frac{t^2}{2!} x^2 + \frac{t^3}{3!} x^3 + \dots\right] \\&= E(1) + t E(x) + \frac{t^2}{2!} E(x^2) + \dots + \frac{t^r}{r!} E(x^r) + \dots \\&= 1 + \frac{t}{1!} \mu_1' + \frac{t^2}{2!} \mu_2' + \dots + \frac{t^r}{r!} \mu_r' + \dots\end{aligned}$$

The coefficient of $\frac{t^r}{r!}$ is μ_r' the r^{th} Moment of x about origin

$$E(x) = \mu_1' = \frac{t^1}{1!} \text{th coefficient}; E(x^2) = \mu_2' = \frac{t^2}{2!} \text{th coefficient}$$

or $\mu_1' = \text{first Moment about origin} = E(x)$

$\mu_2' = \text{2nd Moment about origin} = E(x^2)$

So,

$$\text{Mean} = E(x) = \mu_1' = \frac{t^1}{1!} \text{th coefficient.}$$

$$\text{Variance} = E(x^2) - [E(x)]^2$$

$$\text{Variance} = \mu_2' - (\mu_1')^2.$$

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about origin

Moment generating function, with probability distribution.

① Mgf for :-
Discrete Random variable :-

Let X is a discrete R.V. with probability distribution

X	$p(x)$
x_1	p_1
x_2	p_2
$\vdots$	$\vdots$
x_n	p_n

$$\text{then } M_X(t) = E(e^{tx}) = \sum_{i=1}^n e^{tx_i} p_i = M_0(t)$$

Mgf for :-
② Continuous Random variable :-

Let X is a Continuous R.V. with probability density function

$$f(x), \quad -\infty < x < \infty$$

$$\text{then } M_X(t) = E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} f(x) dx = M_0(t)$$

Example :- Find the m.g.f of Random variable X having the probability density function

$$f(x) = \begin{cases} x & 0 \leq x < 1 \\ 2-x & 1 \leq x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find Mean and variance using MGF

$$\text{Sol}^n | - \\ \underline{\quad} \quad M_x(t) = E(e^{tx})$$

$$= \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

$$= \int_0^1 e^{tx} f(x) dx + \int_1^2 e^{tx} f(x) dx + \int_2^{\infty} e^{tx} f(x) dx$$

$$= \int_0^1 e^{tx} x dx + \int_1^2 e^{tx} (2-x) dx + 0$$

$$= \left[x \frac{e^{tx}}{t} \right]_0^1 - \int_0^1 \frac{e^{tx}}{t} \cdot 1 dx + \left[\frac{e^{tx}}{t} (2-x) \right]_1^2 - \int_1^2 \frac{e^{tx}}{t} (-1) dx$$

$$= \frac{e^t}{t} - \left[\frac{e^{tx}}{t^2} \right]_0^1 + \left[-\frac{e^t}{t} \right] + \left[\frac{e^{tx}}{t^2} \right]_1^2$$

$$= \frac{e^t}{t} - \left(\frac{e^t}{t^2} - \frac{1}{t^2} \right) - \frac{e^t}{t} + \frac{e^{2t}}{t^2} - \frac{e^t}{t^2}$$

$$= \frac{e^{2t}}{t^2} - \frac{2e^t}{t^2} + \frac{1}{t^2} = \frac{(e^t - 1)^2}{t^2}$$

$$M_x(t) = \frac{1}{t^2} \left\{ \left[1 + 2t + \frac{(2t)^2}{2!} + \frac{(2t)^3}{3!} + \dots \right] - 2 \left[1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots \right] + 1 \right\}$$

$$= \frac{1}{t^2} \left\{ t^2 + t^3 + \frac{7}{12} t^4 + \dots \right\} = 1 + t + \frac{7}{12} t^2 + \dots$$

$\mu_1' = \text{Coefficient of } \frac{t^1}{1!} = 1 = E(x) = \text{Mean}$

$\mu_2' = \text{Coefficient of } \frac{t^2}{2!} = 2 \times \frac{7}{12} = 7/6$

$$\therefore \text{Var}(X) = \mu_2' - (\mu_1')^2$$

$$= \frac{7}{6} - 1 = \frac{1}{6}$$

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Q 2 Show that the mgf of random variable X having the probability density function

$$f(x) = \frac{1}{3}, \quad -1 < x < 2$$

$$= 0 \quad \text{elsewhere.}$$

is

$$M(t) = \frac{e^{2t} - e^{-t}}{3t}, \quad t \neq 0.$$

$$= 1, \quad t = 0.$$

Solⁿ - for the DRV.

$$M(t) = E(e^{tx}) = \int_{-1}^2 f(x) e^{tx} dx$$

$$= \int_{-1}^2 \frac{1}{3} e^{tx} dx + \int_2^{\infty} 0 dx.$$

$$= \frac{1}{3} \left[\frac{e^{tx}}{t} \right]_{-1}^2 + 0$$

$$= \frac{1}{3t} (e^{2t} - e^{-t}), \quad t \neq 0.$$

if $t = 0$

$$M(t) = \int_{-1}^2 \frac{1}{3} e^0 dx = \frac{1}{3} \int_{-1}^2 1 dx$$

$$= \frac{1}{3} (x)_{-1}^2 = \frac{3}{3} = 1$$

Hence proved

Qns based on DRV of MGF.

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Ex 3 find the mgf of a random variable X , whose probability function is

$$P(X=x) = p(1-p)^{x-1} \quad x=1, 2, \dots$$

Solⁿ

$$M_X(t) = E(e^{tx})$$

$$= \sum_{x=1}^{\infty} e^{tx} P(X=x)$$

$$= \sum_{x=1}^{\infty} e^{tx} p(1-p)^{x-1}$$

$$= \frac{p}{1-p} \sum_{x=1}^{\infty} e^{tx} (1-p)^x$$

$$= \frac{p}{1-p} \left[\sum_{x=0}^{\infty} [(1-p)e^t]^x - 1 \right]$$

$$= \frac{p}{1-p} \left[\left\{ 1 + (1-p)e^t + [(1-p)e^t]^2 + [(1-p)e^t]^3 + \dots \right\} - 1 \right]$$

(1 + D + D^2 + ...)

$$= (1-D)^{-1}$$

$$= \frac{p}{1-p} \left\{ [1 - (1-p)e^t]^{-1} - 1 \right\}$$

$$M_X(t) = \frac{p}{1-p} \left\{ (1 - qe^t)^{-1} - 1 \right\}$$

Ans

Note These are various points MGF.

- | | |
|----------------------|--|
| ✓ (1) $M_0(t)$ | Moment generating function about origin. |
| (2) $M_a(t)$ | " " " " any point |
| (3) $M_{\bar{x}}(t)$ | " " " " Mean // |